The Sequential Profiled-Gravity Procedure: Methodology for the Divergence-Constrained Upper/Lower Bounds

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Abstract

This note documents, in full technical detail, the procedure used to produce the "main" upper and lower bound estimates for the gains-from-trade robustness exercise (the runs labeled `upper (regular, W=80,000)` and `lower (regular, W=80,000)` in the accompanying output). The method is a divergence-constrained (Christensen–Connault-type) robust-inference calculation in which all bilateral efficiency shifters A_{od} except the destination country's own column are *profiled out* via an inner destination-share inversion, rather than carried as free outer parameters. Gravity's origin×destination fixed-effect orthogonality restriction is then imposed on the resulting reweighted distribution via a *linearized (influence-function)* moment, re-derived and re-imposed every time the loop re-solves. We give: the model and gauge conventions; the exact definition of the destination-share inversion and its Newton solve; the full derivation of the influence function used to linearize the gravity restriction; the augmented moment system fed to the inner divergence-minimization solve; the fixed-point ("sequential") iteration used to drive the exact nonlinear gravity residual to zero; the outer optimization problem over $(\gamma'_{\text{focal}}, A_{\cdot,\text{focal}})$ and precisely how its gradient is computed (including the winner-boundary/Dirac-term correction this required); every convergence check and tolerance used; and what happens, at each level, when something fails to converge.

Contents

1 Motivation: the dimensionality problem

The underlying structural model is a standard Eaton–Kortum-style discrete-choice gravity model with D countries. Each country pair (o, d) has a bilateral efficiency shifter A_{od}. A fully general divergence-constrained robustness exercise would carry *every* A_{od} (D^2 of them, $D^2 - D$ free after a scale normalization) as a free parameter in the outer optimization. This does not scale: at $D = 20$ that is 400 parameters (in fact this is exactly the "full-A-in-outer-loop" comparison method run alongside the sequential method as a cross-check).

The procedure below instead keeps only the focal destination's own column, $A_{\cdot,\text{focal}}$ (D parameters), in the outer optimization. Every other column $A_{\cdot,d}$, $d \neq$ focal, is *recovered* at each outer evaluation by inverting that destination's observed trade-share column against the current candidate reweighted distribution F. Because that inversion is a nonlinear, non-explicit function of F (it is the solution of a fixed-point/root-finding problem, not a closed-form moment), it cannot enter the inner divergence-minimization problem directly as a CC moment. The resolution is to *linearize* the resulting gravity restriction's dependence on F around the current candidate F (an influence-function/Gateaux-derivative argument), add that linearization as one extra scalar CC moment, re-solve, re-invert under the new F, recompute the *exact* nonlinear gravity residual, and iterate until it is within tolerance. This lets gravity genuinely shape which distribution the divergence-minimization problem selects, rather than being checked only after the fact.

2 Model, notation, and gauge

2.1 Draws and prices

Base productivity draws $U_{s,o} \sim \text{Exp}(1)$, i.i.d. across simulation draws $s = 1, \ldots, W$ and origins $o = 1, \ldots, D$ (one draw per *origin*, shared across destinations). Let μ be the Frchet-type shape parameter and σ the (fixed) elasticity of substitution. Define

$$\log x_{s,o} \;=\; \mu(1-\sigma)\log U_{s,o} \;=\; \mu \log(U_{s,o}^{1-\sigma}),$$

which is destination-independent and precomputable once. The level price charged by origin o in destination d on draw s is

$$p_{od}(s) \;=\; w_o \cdot A_{od}^{-1} \cdot \tau_{od} \cdot U_{s,o}^{\mu},$$

where w_o is the wage, $\tau_{od} \geq 1$ is the bilateral iceberg/tariff cost, and A_{od} is the structural efficiency shifter (so the code's internal variable named `AodPow` is A_{od}^{-1}; this is purely a naming artifact of the implementation, not a modeling choice). Define the **competitiveness index**

$$u_{o,d} \;=\; (\sigma-1)\Big(\log A_{od} - \log w_o - \log \tau_{od}\Big).$$

Since $\sigma > 1$, $p_{od}^{1-\sigma}$ is monotone decreasing in p_{od}, so the winning (cheapest) origin on draw s is

$$\text{winner}(s,d) \;=\; \arg\min_o p_{od}(s) \;=\; \arg\max_o \big(u_{o,d} + \log x_{s,o}\big).$$

2.2 Trade shares under a reweighted distribution

Let F be a probability distribution over the W draws with weights $p(s)$ (the "least-favorable distribution," or LFD, that the divergence-minimization problem selects), and let F^* denote the base measure with $p^*(s) = 1/W$. The F-implied trade share of origin o in destination d is

$$\text{share}_d(o;F) \;=\; \frac{\sum_{s:\,\text{winner}(s,d)=o} p(s)\cdot e^{u_{o,d}+\log x_{s,o}}}{\sum_s p(s)\cdot e^{u_{\text{winner}(s,d)},d} + \log x_{s,\text{winner}(s,d)}}.$$

Equivalently, with $V_d(s) = \max_o(u_{o,d} + \log x_{s,o})$ the log of the winning value, $\text{share}_d(o;F) = \sum_{s:\text{winner}(s,d)=o} p(s)e^{V_d(s)} \big/ \sum_s p(s)e^{V_d(s)}$. This is matched, for every destination $d \neq$ focal, against the observed data trade-share column $\hat{\lambda}_{\cdot,d}$.

2.3 Gauge

Trade shares are invariant to adding an arbitrary constant to an entire column $u_{\cdot,d}$ (it cancels in the ratio above). Two gauges appear in this procedure and are **not** the same thing:

- **Destination-inversion gauge**: the Newton solve for each omitted column pins the reference origin, $u_{1,d} = 0$, and solves for the remaining $D-1$ free coordinates.
- **Outer-parameterization gauge**: the outer optimization normalizes $\gamma_{\text{focal}} \equiv 1$ (Section 4), which *replaces* (does not add to) the older $A_{1,\text{focal}} = 1$ pin. All D entries of $A_{\cdot,\text{focal}}$ are free under this normalization.

These two gauges being different conventions is completely immaterial to the trade shares or the reported bounds (both are valid gauge choices and the model is gauge-invariant after two-way demeaning); it is only a hazard if one tries to compare a raw competitiveness matrix against an external closed-form calibration without first correcting for the gauge difference.

3 The robustness objective and the divergence budget

The parameter of interest is a gains-from-trade-type object $\kappa \in [0,1]$. Under the $\gamma_{\text{focal}} \equiv 1$ normalization,

$$\kappa \;=\; 1 - \big(\gamma'_{\text{focal}}\big)^{\sigma/(\sigma-1)},$$

where γ'_{focal} is a (counterfactual) normalization constant appearing directly as one entry of the outer parameter vector θ. This transform is monotone decreasing in γ'_{focal}, so *minimizing* γ'_{focal} over the feasible set gives the **upper** bound on κ and *maximizing* it gives the **lower** bound.

The "feasible set" is a divergence ball around the base measure F^*: the search is only allowed to consider distributions F whose divergence from F^* is at most a budget δ. The divergence itself is the same Christensen–Connault (CC) hybrid exponential/quadratic functional used throughout this codebase's inner solves. Its convex-conjugate representation is

$$\Psi(a) \;=\; \begin{cases} e^a - 1, & a \le 1, \\ \frac{e}{2}\big(a^2+1\big) - 1, & a > 1, \end{cases} \qquad \Psi'(a) = d\Psi(a) = \begin{cases} e^a, & a \le 1, \\ e \cdot a, & a > 1, \end{cases}$$

and the corresponding primal divergence generator (its Legendre dual) is

$$\varphi(m) \;=\; \begin{cases} m\log m - m + 1, & 0 < m \le e, \\ \dfrac{m^2}{2e} - \dfrac{e}{2} + 1, & m > e, \end{cases} \qquad \operatorname{div}(F) \;=\; \frac{1}{W}\sum_{s=1}^{W} \varphi\big(W \cdot p(s)\big).$$

By construction $\operatorname{div}(F^*) = 0$.

4 Reduced parameterization: only $A_{\cdot,\text{focal}}$ is free

The outer parameter vector, for the runs documented here, is

$$\theta \;=\; \big(\mu,\; \sigma,\; \gamma'_{\text{focal}},\; A_{1,\text{focal}}, \ldots, A_{D,\text{focal}}\big),$$

of which μ **and** σ **are held fixed** at their calibrated values throughout the search (the logs record this explicitly as `MU_FIXED(removed)=true`); only γ'_{focal} and the D entries of $A_{\cdot,\text{focal}}$ are free outer-optimization variables ($D+1$ free parameters total, versus $D^2 - D$ under the full-A comparison method).

Under $\gamma_{\text{focal}} \equiv 1$, an exact closed-form autarky argument (using the model's own Frchet aggregator identity for the domestic/focal share, cross-checked independently against an unrelated derivation in this project and confirmed to agree) gives the *theoretically exact* range for γ'_{focal} implied by $\kappa \in [0, \kappa_{\max}]$:

$$\kappa_{\max} \;=\; 1 - \hat{\lambda}_{dd}^{\,1/(\sigma-1)}, \qquad \gamma'_{\text{focal}} \in \big[\hat{\lambda}_{dd}^{\,1/\sigma},\; 1\big],$$

where $\hat{\lambda}_{dd}$ is the observed domestic/own trade share of the focal country. These are used as hard bounds on $\theta_3 = \gamma'_{\text{focal}}$ in the outer KNITRO solve, so the search is structurally prevented from exploring (γ, γ') combinations that could not correspond to any genuine model equilibrium, rather than relying only on catching such violations after the fact.

The *reduced* inner moment system used at every outer evaluation ($D+1$ moments before the gravity linearization is appended) is:

- D **focal trade-share moments**: for each origin $o = 1, \ldots, D$, the (hard-argmax) value contributed by o winning at the focal destination, minus its data-share-implied target, $\hat{\lambda}_{o,\text{focal}} \cdot \text{gdp}_{\text{focal}}$.
- 1 **counterfactual price-index moment**: matches the focal country's own counterfactual (post-shock) price index to the target implied by γ'_{focal} directly. This is the *only* place γ'_{focal} enters the moment system, and it enters *smoothly* (no hard-argmax dependence on it) a fact used below in the outer-gradient computation.

5 Destination-share inversion

For every omitted destination $d \neq$ focal, given the current candidate distribution F (weights p), the procedure recovers $u_{\cdot,d}$ by solving

$$\hat{u}_{\cdot,d} \;=\; \operatorname*{arg\,min}_{u:\,u_1=0} \;\phi_d(u), \qquad \phi_d(u) \;=\; \log\sum_{s=1}^{W} p(s)\exp\Big(\max_o\big(u_o + \log x_{s,o}\big)\Big),$$

whose gradient on the free coordinates is exactly $\nabla\phi_d = \text{share}_d(\cdot; F)$, so the first-order condition $\nabla\phi_d - \hat{\lambda}_{\cdot,d} = 0$ is precisely "match the observed share column." In practice the pipeline solves this at a small positive softmax temperature $\rho = 2\times10^{-3}$ (replacing the hard max with $\rho\log\sum\exp(\cdot/\rho)$), which makes ϕ_d everywhere C^∞ so a damped (Levenberg–Marquardt) Newton method has a well-defined gradient/Hessian to work with; $\rho \to 0$ recovers the literal hard-argmax economic model. Away from ties the closed-form Hessian on the free coordinates is the softmax covariance of the winner indicator,

$$H_d \;=\; \big(\text{diag}(\text{share}_d) - \text{share}_d\,\text{share}_d^\top\big)_{[\text{free},\text{free}]},$$

which is positive semi-definite and generically positive definite on the free coordinates.

Convergence and safeguards. Each destination's Newton solve runs for up to 150 iterations with up to 50 line-search steps per iteration, and is declared converged when the maximum absolute share error is below `DEST_INV_TOL`$= 10^{-6}$ (environment-configurable). If the free coordinates' norm exceeds 10^8 at any point the solve bails out immediately rather than grinding to the iteration cap this catches the genuine structural-degeneracy failure mode where, at extreme θ (e.g. μ far from its calibrated value, or a divergence-implied reweighting that makes a target share numerically unreachable given an ill-conditioned H_d), the Newton iterates run off to infinity. **If any single omitted destination fails to converge, the entire evaluation at that θ is treated as infeasible** (Section 12) a non-converged $u_{\cdot,d}$ is not trustworthy even if it happens to produce a superficially small gravity residual.

6 The gravity identification restriction

With the full competitiveness matrix $u_{o,d}$ assembled (focal column from θ directly; every omitted column from the inversion above), define $\log A_{od} = \log w_o + \log\tau_{od} + u_{o,d}/(\sigma-1)$. Let $Q = \log\tau$ and let $\tilde{Q}$, $\widetilde{\log A}$ denote the two-way (origin- and destination-fixed-effect) demeaned versions of Q and $\log A$ over the *full* $D \times D$ grid (including the diagonal $o = d$, matching the pre-existing gravity-moment convention elsewhere in this codebase). The identifying restriction is the standard gravity orthogonality condition $\mathbb{E}[\Delta\Delta\log A \cdot \Delta\Delta\log\tau] = 0$, whose finite-sample analogue is

$$R_{\text{sum}} \;=\; \sum_{o,d} \tilde{Q}_{od}\cdot\widetilde{\log A}_{od}, \qquad R_{\text{mean}} \;=\; R_{\text{sum}}/D^2, \qquad R_{\text{beta}} \;=\; R_{\text{sum}}\Big/\textstyle\sum_{od}\tilde{Q}^2_{od}.$$

All three vanish together ($R_{\text{sum}} = 0 \iff R_{\text{mean}} = 0 \iff R_{\text{beta}} = 0$), so they are the same identifying condition; they differ only in how a finite numerical tolerance against them behaves. R_{mean} **is the quantity actually checked against tolerance** ($|R_{\text{mean}}| \leq \text{tol} = 5\times10^{-4}$ is the "gravity feasible" gate), because it is the literal, unembellished sample-moment average with no extra variance normalization. R_{beta}'s numerical *scale* (not its role as the tolerance check) is still used for the solver-facing linearized moment and its gradient, because R_{mean}'s much smaller magnitude was empirically found to make the outer search take oversized, destabilizing steps; since $\sum_{od}\tilde{Q}^2_{od}$ is a data-only constant, $R_{\text{beta}} \equiv (D^2/\sum\tilde{Q}^2)\cdot R_{\text{mean}}$ exactly, so this is purely a numerical-conditioning choice, decoupled from the (separately correct) tolerance check.

7 The influence function: linearizing gravity's dependence on F

This is the mathematical core of the procedure: since R depends on F only through the destination inversions $u_{\cdot,d}(F)$, and those inversions are themselves solutions of a nested fixed-point problem (not an explicit function of F), R cannot be written down as, or differentiated as, an ordinary CC moment. The fix is to compute its *Gateaux derivative* at the current F and use that local linear approximation as one additional scalar moment.

7.1 Draw-level objects

At a converged inversion $u_{\cdot,d}$, with $V_d(s) = \max_o(u_{o,d} + \log x_{s,o})$, $\mathrm{Vexp}_d(s) = e^{V_d(s)}$, and $M_d = \sum_s p(s)\,\mathrm{Vexp}_d(s)$, define the free-coordinate score perturbation object

$$\xi_d(s,o) \;=\; \mathrm{Vexp}_d(s)\cdot\big(\mathbf{1}\{\text{winner}(s,d)=o\}-\hat\lambda_{o,d}\big), \qquad o \text{ ranging over the free (non-reference) origins.}$$

By construction $\mathbb{E}_F[\xi_d(\cdot,o)] = 0$ at a converged inversion (this is exactly the first-order condition of the Newton solve).

7.2 Implicit derivative of the inversion

For an arbitrary mean-zero score perturbation h (i.e. a direction in which F could be tilted, with $\mathbb{E}_F[h]=0$), the implicit function theorem applied to the inversion's fixed-point condition gives the Gateaux derivative of $u_{\cdot,d}$ in that direction *without ever forming H_d^{-1} explicitly*:

$$D_F u_d[h] \;=\; -H_d^{-1}\cdot\frac{1}{M_d}\,\mathbb{E}_F\big[h\cdot\xi_{d,\text{free}}\big].$$

7.3 Adjoint solve and the influence function

Let $c_d = \tilde Q_{\cdot,d}/(\sigma-1)$ restricted to the free coordinates (the sensitivity of R to $u_{\cdot,d}$ directly, from the closed-form identity $\widetilde{\log A} = \tilde Q + \tilde u/(\sigma-1)$). Rather than differentiate R through $D_F u_d[h]$ directly (which would require the explicit H_d^{-1}), solve the *adjoint* system

$$H_d\,a_d \;=\; c_d$$

(exploiting H_d's symmetry), and then the pointwise influence function of the gravity residual is

$$\psi_R(s) \;=\; -\sum_{d\neq\text{focal}}\frac{a_d^\top\xi_{d,\text{free}}(s)}{M_d}, \qquad \bar\psi(s) \;=\; \psi_R(s)-\mathbb{E}_F[\psi_R].$$

This satisfies, by construction, $\mathbb{E}_F[\psi_R]=0$ automatically, and the influence of an arbitrary tilt h on R is exactly $\mathbb{E}_F[h\cdot\psi_R]$ i.e. $\bar\psi$ is the score whose CC moment condition $\mathbb{E}_F[\bar\psi]=0$ (or, offset, $\mathbb{E}_F[\bar\psi]=-R$) is the correct first-order surrogate for "move F so that the gravity residual moves by $-R$, i.e. toward zero." *The linearization is recomputed from scratch (not accumulated) at every iteration of the sequential loop below*, since it is only a locally valid, first-order surrogate for the true nonlinear restriction.

7.4 Directional-derivative validation

Because the closed-form H_d used above is the exact Hessian of the finite-sample softmax potential *away from winner ties*, but (being derived from a hard-argmax-differentiated branch) omits the population extensive-margin (boundary-switching) term, this construction is validated against a direct numerical directional derivative (perturb F, re-invert exactly, recompute R, compare to the predicted linear change) before being trusted; this check passed at relative error $\sim 10^{-4}$ on synthetic data and 3.8×10^{-5} on real pipeline draws.

8 The augmented inner (divergence-minimization) problem

At a given outer θ and a given sequential-loop iterate F (weights p), the inner problem is the standard CC dual: find dual variables $x = (\zeta, \lambda_1, \ldots, \lambda_{D+2})$ solving

$$\min_{\zeta,\lambda} \quad \frac{1}{W}\sum_{s=1}^{W} \Psi\Big(- \zeta - \textstyle\sum_j \lambda_j G_j(s;\theta)\Big) + \zeta,$$

whose optimal value is exactly $\delta^*(\theta)$ the minimum divergence from F^* needed to satisfy all $D+2$ equality moments $\mathbb{E}_F[G_j] = 0$ simultaneously by convex (Fenchel) duality. The moment vector $G(\cdot;\theta)$ is:

1. the D focal trade-share moments (Section 4);
2. the 1 counterfactual price-index moment;
3. the 1 linearized-gravity moment $\bar{\psi}_k$, offset so its target is $-R_k$ (subscript k denotes the current sequential-loop iteration; this moment is entirely replaced, not accumulated, each iteration).

Given the optimal dual solution $x^* = (\zeta^*, \lambda^*)$, the recovered LFD weights are

$$m(s) \;=\; d\Psi\big(- \zeta^* - \textstyle\sum_j \lambda_j^* G_j(s;\theta)\big), \qquad p(s) \;=\; m(s) \Big/ \textstyle\sum_r m(r),$$

which is the standard exponential-tilting form for this class of divergence problems, evaluated here on the finite empirical support of W draws under base weight $1/W$.

The KNITRO solver used for this dual optimization reports a termination status code (`nStatus`) for every call; only codes $\{0, -100, -101, -103\}$ (optimal, or feasible-and-terminated-on-a-tight tolerance) are accepted as a genuine solve any other code (in particular -300, "problem determined unbounded," which for this convex-dual construction is the signature of the underlying *primal* moment-matching problem being infeasible) causes the call to be rejected and treated as a failed inner solve.

9 The sequential fixed-point iteration (at a fixed θ)

Putting Sections 5–8 together, at a fixed outer θ:

1. Solve the *reduced* inner problem (focal moments only, $D+1$ of them; no gravity moment yet) to get an initial F_0 (weights p_0).
2. Invert every omitted destination under p_0; assemble u; compute the exact R_0.

3. Iterate $k = 0, 1, 2, \ldots$ (up to maxit = 20):

   (a) If $|R_k| \leq$ tol: **converged**, stop.

   (b) Otherwise compute the influence function $\bar{\psi}_k$ (Section 7) at the current (u, p_k), solve the augmented ($D + 2$-moment) inner problem to get a *candidate* p_{cand}.

   (c) **Damped update with backtracking line search**: for $\alpha \in \{1, \frac{1}{2}, \frac{1}{4}, \ldots\}$ (up to 12 halvings), form $p_\alpha = (1 - \alpha)p_k + \alpha\, p_{\text{cand}}$, re-invert all omitted destinations under p_α (warm-started the very first, $\alpha = 1$, trial warm-starts from u at p_k; subsequent smaller-α trials warm-start by interpolating between the two known endpoint solutions rather than restarting cold, which was found to cut total Newton iterations by roughly 1.7× with a bit-identical accepted trajectory), and accept the first α for which *every* destination converges *and* $|R_\alpha| < |R_k|$. If no α in the schedule satisfies this, stop (see Section 12).

   (d) Set $p_{k+1} = p_\alpha$, $u_{k+1} = u_\alpha$, $R_{k+1} = R_\alpha$; continue.

4. Report the converged $\operatorname{div}(F_{\text{final}})$ (computed via the *exact primal* divergence functional φ applied directly to the accepted weights necessary because a damped convex combination p_α is *not* itself the argmin of any single divergence-minimization problem, so its divergence cannot simply be read off a solver's internal value) as $\Delta_{\text{sequential}}(\theta)$, and declare the θ-evaluation **gravity-feasible** iff $|R_{\text{final}}| \leq$ tol **and** $\Delta_{\text{sequential}}(\theta) \leq \delta$ *both* conditions are required (see Section 12 for why conflating them was a real, previously-fixed bug).

10 The outer optimization problem

The outer problem, solved by KNITRO, is:

$$\min_{\theta} \ \text{(or } \max_{\theta}\text{)} \ \ \gamma'_{\text{focal}} \qquad \text{s.t.} \qquad \Delta_{\text{sequential}}(\theta) \ \leq \ \delta, \qquad \gamma'_{\text{focal}} \in \big[\hat{\lambda}_{dd}^{1/\sigma},\, 1\big],$$

with μ, σ fixed and $A_{\cdot,\text{focal}}$ otherwise unconstrained (generic wide bounds). Minimizing γ'_{focal} gives κ_{upper}; maximizing gives κ_{lower}. **Gravity does not enter as a second outer constraint** in this method it is folded entirely into $\Delta_{\text{sequential}}(\theta)$ via the sequential loop's own inner augmented moment (Section 9). This is a deliberate design choice whose validity is checked directly (Section 10.1) it is what allows the divergence-budget-constraint gradient to be computed as a single scalar envelope-theorem derivative, with no additional total-derivative correction for a second constraint.

10.1 The outer gradient

Two gradients are needed for KNITRO's SQP steps.

Objective gradient. Trivial: $\gamma'_{\text{focal}} = \theta_3$ enters the objective directly, so its gradient is the unit vector e_1 (in the free-coordinate ordering $(\gamma'_{\text{focal}}, A_{1,\text{focal}}, \ldots, A_{D,\text{focal}})$), sign-flipped depending on whether the search is minimizing or maximizing.

Constraint gradient (the hard part). At the optimal dual solution (ζ^*, λ^*) of the inner problem, the *envelope theorem* says the total derivative of $\Delta_{\text{sequential}}(\theta)$ with respect to θ equals the *partial* derivative of the dual objective with respect to θ, holding (ζ^*, λ^*) fixed because the inner problem's own first-order conditions in (ζ, λ) vanish at the optimum. Concretely, this is the scalar function

$$Q(\theta) \;=\; \frac{1}{W} \sum_{s=1}^{W} d\Psi(a_0(s)) \cdot \sum_j \lambda_j^* \, G_j(s;\theta),$$

(with $a_0(s)$ evaluated at the converged solution), whose gradient in θ is the constraint gradient KNITRO needs.

This is exact and unproblematic for $\gamma'_{\mathbf{focal}}$: γ'_{focal} enters G only through the smooth counterfactual price-index moment, never through any hard-argmax winner rule, so plain automatic differentiation of Q with respect to θ_3 is exact. It is **not** unproblematic for the $A_{\cdot,\text{focal}}$ block: those parameters directly enter the focal destination's own winner rule (Section 2), a hard $\arg\max$ whose derivative is zero almost everywhere and undefined at ties. Naive automatic differentiation through this hard $\arg\max$ silently **drops the winner-boundary (extensive-margin, Dirac-mass) term** the contribution from draws that switch which origin wins as $A_{\cdot,\text{focal}}$ moves and this was confirmed, not merely suspected: a validated diagnostic comparing the naive-AD gradient against a from-scratch finite-difference gradient (which genuinely re-evaluates the winner rule at perturbed points and so captures real winner switches) found a real, non-negligible discrepancy.

The correction used for the reported bounds (`gradient_method=fixed_dual_fd_full`, the default for these runs): the $A_{\cdot,\text{focal}}$ block of the constraint gradient is instead computed by *central finite differences directly on* $Q(\theta)$, perturbing each coordinate of $A_{\cdot,\text{focal}}$ multiplicatively in log-space by a step $h = 0.1$, with the dual variables (ζ^*, λ^*) held fixed at their already-converged value (still an envelope-theorem argument only the derivative-taking method for the θ-dependence changes, from automatic differentiation to finite differences) but the moment matrix G *genuinely recomputed* at each perturbed point, so any draws whose winner switches within the finite step are correctly reflected. This directly captures the winner-boundary term that automatic differentiation cannot see. (Two earlier, now-deprecated variants, `:fixed_dual_fd` and `:boundary`, attempted an *additive correction* to the plain-AD gradient on a reduced $(D+1)$-moment problem; this was found not to be generally valid, because the gravity moment's own dual multiplier λ_R enters the same nonlinear $\Psi(\cdot)$ argument as every other moment, so its contribution to the winner-boundary jump cannot be isolated and dropped. The full, $(D+2)$-moment, direct-replacement method described above is the one actually used for the reported results.)

10.2 Known imprecision this does *not* introduce

The augmented moment's own dependence on θ, when the inner solver's automatic-differentiation machinery needs it (a separate need from the outer KNITRO gradient above this occurs inside the inner solve's own handling of dual/derivative number types), is represented by a *frozen local linear surrogate* using the same envelope-theorem sensitivity of R to θ (holding F fixed) used to build the influence-function moment itself, rather than differentiated through the discrete inversions directly. This affects only the *efficiency* of the search (how directly it is steered toward the optimum), never the validity of a reported bound: feasibility of every reported point is always checked *exactly* (the nonlinear R and the exact primal divergence functional, never the linear surrogate), so a reported bound can only be as good, never systematically biased, regardless of how good the gradient guidance was along the way.

11 Checks, validation, and multistart

- **Gravity feasibility**: $|R_{\text{mean}}| \leq 5 \times 10^{-4}$ at the converged sequential-loop iterate.

- **Divergence-budget feasibility**: $\mathrm{div}(F_{\mathrm{final}}) \leq \delta$, checked via the *exact primal* functional φ applied directly to the accepted (possibly damped-combination) weights, not read off any solver's internal value. **Both** of the above are required for a θ to count as feasible; requiring only the first was a real, previously-diagnosed bug (it let the search report economically impossible negative κ, since the theory proves $\kappa \in [0, \kappa_{\max}]$).

- **Destination-inversion convergence**: every omitted destination's own Newton solve must converge (max share error $< 10^{-6}$) at every iterate used; a non-converged destination invalidates the whole evaluation (Section 5).

- **Best-feasible-θ tracking**: KNITRO's own terminal iterate at the outer iteration cap is not always the best (or even a feasible) point the search visited along the way the search can wander off a good point near the cap. The procedure therefore separately tracks the best feasible θ (extremal κ subject to feasibility) seen across the *entire* search and reports it alongside KNITRO's raw terminal answer (`best_feasible_kappa`/`best_feasible_gp` versus the raw `kappa`/`gamma_p` columns). Because the sequential loop is warm-started from the previous θ's state (not a pure function of θ alone), the tracker snapshots the destination-inversion state that achieved feasibility and re-verifies *warm*, not cold, to avoid a spurious failure on re-check.

- **Hard-max (rho→0) verification**: a fully independent, after-the-fact check given a converged production point's (θ, u, p), re-invert every omitted destination's competitiveness column via a ρ-continuation homotopy down to the literal zero-temperature hard-argmax model (not the $\rho = 2 \times 10^{-3}$ softmax used throughout the main solve), and check that the resulting hard-max gravity residual is within the same tolerance. This is a *separate* flag (`hardmax_verified`) from the smoothed-model `gravity_ok` / `best_feasible_gravity_ok` used to select the reported point; the two can disagree (a bounded, nonzero hard-max share-matching error at higher δ is expected and is not itself evidence of a problem; it does not gate or change `best_feasible_kappa`).

- **Five-start multistart, per δ**: the outer search is run from five different starting points (one anchored at the closed-form calibration, A^*; four random perturbations) at every δ on the grid, to guard against the outer KNITRO search converging to a local/non-global point or hitting its iteration cap short of the true extremum; the reported number at each δ is the best across all five starts. Within a given start, each δ on the grid is warm-started from that same start's solution at the previous δ.

12 Failure modes: what happens when it does not converge

- **A single destination's Newton solve diverges** (competitiveness norm exceeds 10^8, or fails to reach the 10^{-6} share-error tolerance within 150 iterations): the whole evaluation at that θ (and that iterate) is marked infeasible; a plausible genuine cause is μ moving far from its calibrated value, making origins nearly tied and requiring increasingly extreme competitiveness values to match a fixed target share this can be genuine economic/numerical infeasibility, not merely a bug.

- **The backtracking line search in the sequential loop exhausts all** 12 **halvings** without finding an α that both keeps every destination converged and reduces $|R|$: the loop stops where it is (not converged), and the θ-evaluation is reported as gravity-infeasible.

- **The sequential loop reaches its iteration cap (20) without $|R|$ falling below tolerance**: same outcome reported infeasible at that θ.
- **Consequence for the outer search, in every case above**: the moment closure feeds the outer KNITRO problem a deliberately *unsatisfiable* constant column in place of the (missing) gravity moment a strictly positive, non-constant sentinel vector for which $\mathbb{E}_F[\cdot] = 0$ is impossible for *any* distribution F. This makes the inner divergence-minimization problem provably infeasible at that θ, so KNITRO's own outer solve correctly treats that θ as infeasible and moves away from it. (The alternative of silently reusing the previous iterate's gravity column was tried historically and is explicitly wrong: it lets the search escape to gravity-violating θ that only *looks* fine because a stale, no-longer-applicable moment was reused.)
- **KNITRO's own inner-solve status code is anything other than** $\{0, -100, -101, -103\}$ (in particular -300, unbounded-dual/infeasible-primal): that solve is rejected outright and treated as a failed evaluation, regardless of whether it returned superficially finite-looking numbers.

13 Practical settings used for the reported bounds

Setting	Value
Softmax temperature ρ (destination inversion)	2×10^{-3}
Destination-inversion tolerance (`DEST_INV_TOL`)	10^{-6}
Destination-inversion iteration cap / line-search cap	150 / 50
Sequential-loop gravity tolerance (on R_{mean})	5×10^{-4}
Sequential-loop iteration cap	20
Backtracking line-search schedule	$\alpha \in \{1, \frac{1}{2}, \frac{1}{4}, \dots\}$, up to 12 halvings
Outer gradient method	`fixed_dual_fd_full` (Section 10.1)
Outer per-coordinate variable scaling	on (`use_var_scaling`, power = 1.0)
μ, σ	fixed at calibrated values (not searched)
Multistart	5 starts (A^*-anchored + 4 random) per δ
δ grid	0.1, 1.0, 2.0, 5.0
Monte Carlo draws W	80,000

The per-coordinate KNITRO variable scaling deserves a note: a probe evaluation at the initial θ found the constraint-gradient magnitude for γ'_{focal} to be many orders of magnitude larger than for the $A_{\cdot,\text{focal}}$ block, which otherwise left the A block effectively unexplored by the search regardless of gradient method. The fix rescales only the A block's KNITRO-internal variable scale, relative to γ'_{focal}'s own natural constraint-gradient magnitude (leaving γ'_{focal}'s own scale at 1.0, since also rescaling it was found to shrink the *objective's* gradient toward zero and caused the search to falsely satisfy its own KKT stopping test at the initial point without moving at all).